

DETERMINATION OF RELIABILITY WORTH FOR DISTRIBUTION SYSTEM PLANNING

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Abstract: There is an ever growing demand to justify reliability projects on a more quantifiable basis, i.e. investments related to the provision of service reliability needs to be carefully evaluated in regard to their cost and benefit implications. In addition, utilities are recognizing the significant customer and community costs that are incurred when electric supply is abruptly curtailed. Consequently, the utility industry is under pressure to assess their reliability evaluation tools and formulate methods for incorporating the economics of reliability in the decision-making process. This paper is concerned with the evaluation of a reliability worth index that can be used to make decisions in distribution system planning and design. The reliability worth index is termed the interrupted energy assessment rate (IEAR) and is obtained by relating the reliability indices to the customer cost of interruption data. Three fundamentally different approaches for evaluating distribution system reliability worth indices are presented and compared using a small but comprehensive test system. The impact on the worth indices of distribution system operating policies and configurations is also illustrated using the three methods. The worth indices developed can be used to determine the monetary implications of customer supply interruptions thereby incorporating economics in the reliability cost-benefit equation.

Keywords: Customer interruption costs, interrupted energy assessment rate, cost/worth analysis, value-based reliability.

1. INTRODUCTION

The basic function of an electric power system is to meet its energy and load demand at the lowest possible cost to its customers while maintaining acceptable levels of quality and continuity of supply. What constitutes an 'acceptable' level can be examined in terms of the costs and the worth to the consumer of providing an adequate supply. This fact is now widely recognised and efforts are being made to quantify the worth of electric service reliability [1,2,3,4,5]. A complete power system can be divided into the functional zones of generation, transmission and distribution which can be combined to form three hierarchical levels, namely HLI, HLII

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and HLIII [6]. The loss of energy expectation or expected energy not supplied (EENS) has been used in conjunction with an interrupted energy assessment rate (IEAR) to estimate future interruption costs associated with power system deficiencies. The system IEAR is a factor that defines the cost to a representative consumer of each unit of unsupplied energy due to power interruptions and is a useful index for making decisions related to system reliability. IEAR have been developed for generating systems (HLI) [7], composite systems (HLII) [8], overall electric power systems (HLIII) [9] and distribution systems [10]. The basic objective of this paper is to compare three fundamentally different methods for determining the worth index IEAR for individual customer load points and for the overall distribution system. The methods are presented, illustrated and compared by application to two distribution systems associated with a test system designated the RBTS [11, 12]. The customer IEAR can be utilized in conjunction with the system reliability (adequacy) indices in order to perform value-based reliability analysis and justify new system investments.

A variety of approaches have been used to determine the actual or perceived costs of customer interruptions. One method which has been used to establish acceptable reliability worth estimates is to survey electrical consumers in order to determine the monetary losses associated with supply interruptions. The data compiled from these surveys is used to generate sector customer damage functions (SCDF) and composite customer damage functions (CCDF). The cost of interruption data in \$/kW of peak demand for seven sectors using a 1988 Cdn \$ base is given in Reference [11]. This paper utilizes both the SCDF and the CCDF as the cost functions in relating the load point reliability indices to the worth of service reliability.

The first of the three methods presented in this paper for IEAR evaluation [10] enumerates the various component outage states in order to obtain the cost-related customer load point indices. This method is termed the contingency enumeration method (CEM). The primary customer load point adequacy indices are the average failure rate (α), the average outage duration (r) and the annual outage duration (U) [13]. Additional indices, termed system indices, which are weighted averages of the load point indices are also frequently calculated. The most common system indices are the SAIDI, SAIFI, CAIDI, etc [13]. This paper also describes two methods which utilize these two sets of indices in the determination of the sector IEAR. The average load point indices (α , r and U) in conjunction with the SCDF are utilized in the basic indices method (BIM) whereas the system indices (SAIFI, SAIDI and CAIDI) in conjunction with the composite cost function (CCDF) are utilized in the system indices method (SIM). The CEM, BIM and SIM are compared using the IEAR obtained.

The reliability performance of the test systems [12] is evaluated and compared using all three methods for different operating procedures, protection schemes and modes of restoration. A complete list of all abbreviations used in this paper together with a brief description is presented in the Appendix.

II. CONTINGENCY ENUMERATION METHOD (CEM)

The estimation of IEAR indices in a distribution system requires the cost model, the load model and the system model. The cost model is the sector type and its associated SCDF. The average loads connected at each customer load point comprise the load model. The system model consists of relevant reliability parameters of all components such as the main and lateral feeders, factors such as the inclusion or not of disconnects on the main feeders, fuses in the lateral sections, alternate back-fed supply, replacing a failed low voltage transformer with a spare instead of repairing it, main and/or lateral sections consisting of underground cables or overhead lines (OHL), etc. The following procedure is used to estimate the unsupplied energy EENS, the cost of interruptions ECOST and the IEAR at each system load point.

1. Evaluate, for each contingency j , the load point EENS and ECOST using the following equations:

$$ECOST_{j,p} = C_{j,p} L_{av,p} \lambda_j$$

$$EENS_{j,p} = L_{av,p} U_{j,p}$$

where, $L_{av,p}$ is the average connected load at load point p , $C_{j,p}$ (in \$/kW) is the cost of interruption using SCDF of p , and $U_{j,p}$ and λ_j are the unavailability and failure rate respectively of contingency j .

2. Evaluate the load point EENS, ECOST and IEAR using the following:

$$ECOST_p = \sum_{j=1}^N ECOST_{j,p} \quad (MW-\$/kW-yr) \quad (1)$$

$$EENS_p = \sum_{j=1}^N EENS_{j,p} \quad (MWh/yr) \quad (2)$$

$$IEAR_p = \frac{ECOST_p}{EENS_p} \quad (\$/kWh) \quad (3)$$

where N is the number of outage events that isolate p .

III. BASIC INDICES METHOD (BIM)

This method utilizes the overall load point adequacy indices (λ , r and U) as opposed to using the indices associated with every possible load point isolation event. The cost model required is the SCDF. The overall average outage duration index r_p at load point p is used to obtain the cost c_p in \$/kW from the SCDF of the customer type at load point p . The average load curtailed is the average load connected ($L_{av,p}$). The load point cost-related indices can then be as follows:

$$EENS_p = L_{av,p} \lambda_p r_p \quad (4)$$

$$ECOST_p = L_{av,p} \lambda_p c_p \quad (5)$$

$$IEAR_p = \frac{ECOST_p}{EENS_p} = \frac{c_p}{r_p} \quad (\$/kWh) \quad (6)$$

IV. SYSTEM INDICES METHOD (SIM)

This method utilizes the system performance indices which are weighted averages of the individual load point indices and the number of customers connected [13]. The CCDF is used as the cost model, i.e. the individual sectors are weighted to form the composite customer group for the service area under consideration. The system CAIDI represents the average duration of outage per system customer interruption, and it can be used to estimate the cost in \$/kW from the CCDF.

$$\text{System } EENS = (SAIFI) \cdot (L_{av,bus}) \cdot (CAIDI) \quad (7)$$

$$\text{System } ECOST = (SAIFI) \cdot (L_{av,bus}) \cdot (CAIDI \text{ cost}) \quad (8)$$

$$\text{System } IEAR = \frac{CAIDI \text{ cost}}{CAIDI} \quad (\$/kWh) \quad (9)$$

Equations (6) and (9) are fundamentally identical and simply require the average outage duration and a cost look-up table for evaluating the IEAR values. This is a basic advantage of the basic indices and system indices methods over the CEM.

V. APPLICATION TO THE RBTS

The RBTS is an educational test system for which the complete generation, transmission and customer outage cost data are given in Reference 11. Detailed system data of the distribution systems are given in Reference 12. In order to provide continuity with the cost analysis conducted in this paper, the distribution systems associated with the RBTS buses 2 and 4 are shown in Figures 1 and 2 respectively. Bus 2 has 22 load points with an average load of 12.291 MW comprising residential (Resid.), commercial (Comm.), small industrial (Sm.Ind.) and Government and Institutional (G&I) consumers. Bus 4 has 38 customer points with an average connected load of 24.58 MW comprising residential, commercial and small industrial customers. It must be emphasized here that while the test system is sufficiently small so as to permit the conduct of a large number of reliability studies, it is sufficiently detailed to reflect the actual complexities involved in a practical reliability analyses.

Application of the CEM

The contingency enumeration procedure was applied to the two networks for two basic cases where the entire system is composed of either underground cables or overhead lines (OHL). The case study considered is the basic system with disconnects, with fully available alternate supplies, with fully reliable fusegear in laterals and with low voltage transformer replacement in the case of outage [12]. The ECOST, EENS and IEAR indices evaluated using Equations (1), (2) and (3) respectively are shown in Table 1 for the bus 2. The overall system IEAR is obtained by weighting the customer IEAR with the per unit average load connected and summing all the weighted values. The indices for bus 4 can be obtained in a similar manner and are shown in Table 2.

The unsupplied energy and interruption costs of any customer sector can be aggregated to obtain the sector IEAR. These values are shown in Table 3 for the two networks. At bus 2, for

the cable case the commercial (Comm.) sector has the highest IEAR whereas the G&I and Small industrial users' IEAR are quite comparable. In the case of OHL network, however, the G&I and industrial users' IEAR are quite different, with commercial sector suffering the most in terms of outage costs and IEAR. Bus 4 has the highest IEAR associated with the commercial customers for both OHL and cable cases. The residential (Resid.) and small industrial (Sm.Ind.) users have very different IEAR values for both OHL and cable cases. For both test systems, the residential sector is least affected in terms of ECOST and IEAR. For outages of large durations, the commercial sector is affected most and the residential sector the least, with the industrial and G&I users falling in between the two extremes. The basic objective of this paper is to quantify the losses in terms of a worth index (IEAR) that can be used to predict interruption costs for prospective systems in a given service area. It should be clearly appreciated that any change in system topology, configuration and operating philosophy could directly affect the results. This is illustrated later in this paper by application to the test systems.

Application of the BIM

Table 4 presents the results of a case study on the RBTS bus 2 assuming the feeders to be composed of underground cables. Table 4 gives the basic indices of λ , r and U for each customer load point together with the outage cost in \$/kW and the IEAR. A comparison of the individual load point IEAR obtained using the BIM and the CEM for this case is also presented in Table 4. The last column of Table 4 shows the absolute percentage difference between the CEM and the BIM values with respect to the CEM results. The results clearly show that, in general, the load point IEAR are not comparable, with CEM values higher than their BIM counterparts. The CEM and BIM IEAR for load points 8 and 9 (small industrial users), however, are quite comparable. This is primarily due to the fact that the low voltage transformer is not included in the laterals serving these customers (since they are assumed to be the customer property). Notice also from Tables 1 and 3 that the EENS values remain the same; it is the expected cost of outages that is the main contributing factor in the difference in the IEAR values. The average outage duration used in the BIM is an average of several individual outage durations, and could therefore give results that vary from those provided by the CEM. In general, therefore, the BIM can provide reasonable approximations to the exact contingency enumeration procedure, particularly if the low voltage transformer replacement times are much shorter and thus comparable to the main and lateral feeder outage durations.

Application of the SIM

This method can be used to evaluate overall system IEAR. It cannot, however, be used to obtain the individual load point IEAR. The system indices for the test systems are given in Table 5. The SCDF are weighted by the sector peak demand/energy requirements and the CCDF obtained for the service area are given in Table 6. Utilizing the data given in Tables 5 and 6 and Equations (7), (8) and (9), the calculated EENS, ECOST and IEAR are given in Table 7. A comparison of these values with those obtained using the CEM is also shown in Table 7. It can be seen from this table that the SIM

does not provide good estimates of the system IEAR for the cable cases, but provides comparable results for the OHL cases. This is due to the long outage durations (30 hrs) in the case of cables, as opposed to 5 hrs in the case of OHL systems. As in the case of the BIM, the ECOST and therefore the bus IEAR values obtained using the SIM are lower than those obtained by the CEM. The SIM may therefore be utilized to obtain representative IEAR particularly in cases where the outage durations are short.

VI. COMPARISON OF THE THREE METHODS

The weighted bus IEAR using the BIM, SIM and CEM are compared for the test systems in Table 8. The values in parenthesis in Table 8 are the percentage differences between the CEM and the BIM/SIM with respect to the CEM. The results clearly indicate that the IEAR values using the BIM and the SIM are quite comparable to those from the CEM for the OHL cases. It can also be seen that the BIM and the SIM provide comparable results between the two. It can therefore be concluded that the BIM and the SIM can provide reasonably good estimates of the system IEAR but that a detailed contingency enumeration technique should be used in order to obtain exact estimates. The main advantage of the BIM/SIM is that if the average outage duration at a given customer load point and its associated cost function are known, the IEAR can be readily evaluated using Equation (6), and if the average system CAIDI and the CCDF are available, then the system IEAR can be easily obtained using Equation (9).

Impact of Distribution System Configurations

The customer and sector IEAR for the test systems were calculated in the preceding sections of this paper using three methods. The distribution configuration used was with disconnects, fuses, alternative supply and low voltage transformer replacement in case of failure. This section investigates the variation in sector IEAR due to different operating philosophies and configurations of the radial systems. Six different cases were investigated:

1. Case A : Disconnects; fuses; alternate supply; transformer repair.
2. Case B : No disconnects; no fuses; no alternative supply; transformer repair.
3. Case C : No disconnects; fuses; no alternate supply; transformer repair.
4. Case D : Disconnects; no fuses; alternative supply; transformer repair.
5. Case E (Base case): Disconnects; fuses; alternative supply; transformer replacement.
6. Case F : Disconnects; no fuses; no alternative supply; transformer repair.

Table 9 gives a summary of the bus IEAR for all the cases (including the base case) using the three methods. The system indices required for the SIM are given in Reference 12. The variation in the customer IEAR for these six cases using the CEM for the bus 2 is shown in Figure 3. It can be seen from Table 9 and Figure 3 that for a specified bus and for radial feeders to be composed of one of OHL or cable, the IEAR for cases A to D and case F are quite comparable. The IEAR for case E (base case), however, are quite different than those for

the other five cases. That the low voltage transformer repair significantly increases the sector IEAR should therefore be given due consideration in conducting reliability cost/reliability worth studies in distribution systems.

A pictorial representation comparing the distribution system IEAR using the three methods (CEM, BIM and SIM) is shown in Figure 4 for the buses 2 and 4 (OHL and cable cases). Figure 4 clearly shows that for bus 2 OHL case, the three methods provide comparable results for configurations B and E only. The BIM and the SIM between themselves, however, provide comparable results for most cases. The results for bus 4 can be explained in a similar manner. All three methods provide fairly comparable IEAR for bus 4 for configurations B and E. In general it can be seen from the bar charts of Figure 4 that the BIM and the SIM compare well to one another but are configuration sensitive when compared with the CEM.

VII. CONCLUSIONS

This paper presents the basic concepts involved in utilizing three techniques for evaluating distribution system reliability worth index designated IEAR. The techniques are the contingency enumeration method (CEM), the basic indices method (BIM), and the system indices method (SIM). The customer IEAR are utilized to estimate combined IEAR indices for different sectors associated with two test systems. The CEM is used as the reference method since it makes use of each outage contingency and evaluates the cost of interruption to prospective customers due to each of the outage events.

A comparison of the three methods using the test systems provides results which are quite comparable for some system configurations and quite different for other system configurations. In particular, long duration outages cause large errors in evaluating the average indices using the BIM/SIM. It is therefore suggested that the BIM and the SIM are very sensitive to specific operating policies and may therefore be used selectively for evaluating the worth of electric service reliability. It must be emphasized, however, that these two methods are much simpler and computationally more efficient to apply as compared to the CEM. The load point IEAR or sector IEAR can be used as a single available index in evaluating the worth of electric service reliability in terms of user interruption costs [14].

VIII. REFERENCES

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IX. BIOGRAPHIES

Lalit Goel was born in New Delhi, India in 1960. Obtained a B. Tech. degree in Electrical Engineering from Regional Engineering College, Warangal, India in 1983. Worked in India between 1983 and 1988 for Engineers India Limited, a premier consultancy organization. Obtained his MSc and PhD degrees from the University of Saskatchewan in 1988 and 1991 respectively. Main research interests are power system analysis and cost/benefit reliability evaluation. Presently a Lecturer in the School of Electrical and Electronic Engineering at the Nanyang Technological University in Singapore.

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X. APPENDIX

List of Abbreviations.

EENS	expected unsupplied energy due to power outages
ECOST	expected outage cost to customers due to supply outages
IEAR	interrupted energy assessment rate in interruption\$/kWh not supplied
CEM	contingency enumeration method
BIM	basic indices method
SIM	system indices method
λ	average failure rate (1/yr)
τ	average outage duration (h)
U	average annual outage time (h/yr)
SAIFI	system average interruption frequency index, interruptions/customer-yr
SAIDI	system average interruption duration index, h/customer-yr
CAIDI	customer average interruption duration index, h/customer-interruption
SCDF	sector customer damage function, interruption cost vs duration for sectors
CCDF	composite customer damage function, interruption cost vs duration for a service area

Table 1 Bus 2: Cost-Related Customer Load Point Indices.

Load Pt.	Cust. class	EENS (MWh/yr)	OHL Case ECOST (1000\$/yr)	IEAR (\$/kWh)	EENS (MWh/yr)	Cable Case ECOST (1000\$/yr)	IEAR (\$/kWh)
1	Resid.	0.388	0.533	1.372	1.082	4.264	3.942
2	Resid.	0.423	0.581	1.373	1.210	4.859	4.015
3	Resid.	0.423	0.581	1.373	1.210	4.859	4.015
4	G&I	0.410	0.906	2.207	1.145	10.782	9.421
5	G&I	0.447	0.977	2.184	1.280	12.329	9.630
6	Comm.	0.351	3.148	8.960	1.000	15.807	15.812
7	Comm.	0.341	3.059	8.969	0.953	14.898	15.626
8	Sm.Ind.	0.543	3.631	6.689	1.932	16.292	8.433
9	Sm.Ind.	0.579	3.912	6.753	2.036	17.118	8.410
10	Resid.	0.390	0.533	1.368	1.085	4.268	3.933
11	Resid.	0.423	0.581	1.374	1.210	4.863	4.019
12	Resid.	0.363	0.499	1.374	1.045	4.216	4.035
13	G&I	0.418	0.914	2.186	1.158	10.735	9.270
14	G&I	0.427	0.931	2.181	1.192	11.124	9.333
15	Comm.	0.331	2.974	8.993	0.921	14.404	15.645
16	Comm.	0.359	3.212	8.952	1.027	16.291	15.863
17	Resid.	0.334	0.458	1.373	0.937	3.716	3.966
18	Resid.	0.328	0.449	1.368	0.913	3.590	3.933
19	Resid.	0.357	0.489	1.369	1.021	4.090	4.007
20	G&I	0.449	0.979	2.181	1.284	12.334	9.608
21	G&I	0.418	0.914	2.186	1.158	10.735	9.270
22	Comm.	0.343	3.072	8.967	0.956	14.920	15.604
Total		8.844	33.331	3.936	25.753	216.493	8.429

Table 2 Bus 4: Cost-Related Customer Load Point Indices.

Load Pt.	Cust. class	EENS (MWh/yr)	OHL Case ECOST (1000\$/yr)	IEAR (\$/kWh)	EENS (MWh/yr)	Cable Case ECOST (1000\$/yr)	IEAR (\$/kWh)
1	Resid.	0.319	0.380	1.191	0.716	2.103	2.936
2	Resid.	0.346	0.417	1.205	0.814	2.556	3.139
3	Resid.	0.319	0.380	1.191	0.716	2.103	2.936
4	Resid.	0.354	0.429	1.210	0.847	2.708	3.198
5	Resid.	0.317	0.382	1.205	0.747	2.345	3.139
6	Comm.	0.270	2.437	9.027	0.645	8.890	13.784
7	Comm.	0.263	2.380	9.042	0.620	8.454	13.635
8	Sm.Ind.	0.338	2.567	7.596	0.984	7.910	8.038
9	Sm.Ind.	0.580	4.327	7.458	1.746	14.159	8.109
10	Sm.Ind.	0.403	2.990	7.420	1.224	9.949	8.128
11	Resid.	0.349	0.426	1.221	0.837	2.698	3.223
12	Resid.	0.340	0.414	1.217	0.804	2.546	3.165
13	Resid.	0.340	0.414	1.217	0.804	2.546	3.165
14	Resid.	0.288	0.346	1.203	0.648	1.920	2.963
15	Resid.	0.312	0.380	1.217	0.738	2.336	3.165
16	Comm.	0.259	2.345	9.049	0.613	8.394	13.704
17	Comm.	0.239	2.175	9.102	0.538	7.092	13.187
18	Resid.	0.349	0.418	1.198	0.821	2.563	3.122
19	Resid.	0.323	0.382	1.183	0.723	2.109	2.919
20	Resid.	0.349	0.418	1.198	0.821	2.563	3.122
21	Resid.	0.349	0.418	1.198	0.821	2.563	3.122
22	Resid.	0.296	0.350	1.183	0.663	1.935	2.919
23	Resid.	0.320	0.384	1.198	0.753	2.351	3.122
24	Comm.	0.266	2.403	9.037	0.625	8.494	13.590
25	Comm.	0.246	2.233	9.087	0.550	7.192	13.069
26	Sm.Ind.	0.384	2.855	7.444	1.158	9.398	8.116
27	Sm.Ind.	0.400	2.961	7.406	1.218	9.908	8.135
28	Sm.Ind.	0.335	2.538	7.581	0.978	7.869	8.046
29	Sm.Ind.	0.348	2.656	7.637	1.002	8.032	8.016
30	Sm.Ind.	0.396	2.973	7.498	1.182	9.561	8.089
31	Sm.Ind.	0.522	3.984	7.637	1.503	12.048	8.016
32	Resid.	0.351	0.427	1.217	0.840	2.702	3.215
33	Resid.	0.351	0.427	1.217	0.840	2.702	3.215
34	Resid.	0.316	0.378	1.199	0.710	2.096	2.954
35	Resid.	0.351	0.427	1.217	0.840	2.702	3.215
36	Resid.	0.289	0.347	1.199	0.651	1.923	2.954
37	Resid.	0.322	0.392	1.217	0.771	2.478	3.215
38	Comm.	0.240	2.186	9.099	0.540	7.112	13.163
Total		12.740	52.747	4.704	32.052	197.010	6.346

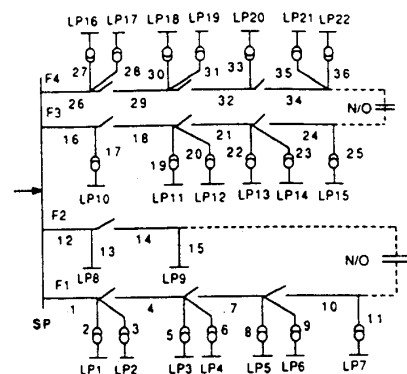


Figure 1 RBTS Bus 2 Radial Distribution System.

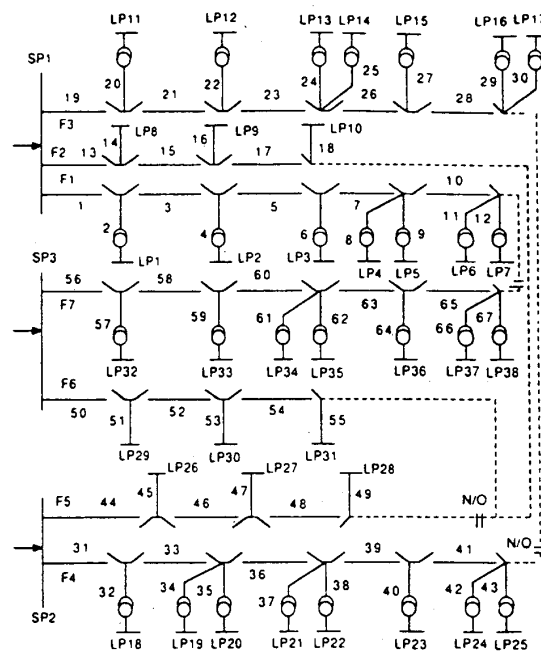


Figure 2 RBTS Bus 4 Radial Distribution System.

Table 3 Buses 2&4: Cost-Related User Sector Indices.

Sector type	EENS (MWh/yr)	OHL Case ECOST (1000\$/yr)	IEAR (\$/kWh)	EENS (MWh/yr)	Cable Case ECOST (1000\$/yr)	IEAR (\$/kWh)
BUS 2:						
Sm.Ind.	1.122	7.543	6.72	3.968	33.41	8.42
Comm.	1.725	15.465	8.96	4.857	76.32	15.71
Resid.	3.429	4.704	1.37	9.713	38.725	3.99
G&I	2.569	5.621	2.19	7.217	68.039	9.43
Total:	8.844	33.331	3.94	25.753	216.493	8.43
BUS 4:						
Sm.Ind.	7.252	8.738	1.20	16.926	52.547	3.10
Comm.	1.783	16.159	9.06	4.131	55.628	13.47
Sm.Ind.	3.705	27.850	7.52	10.995	88.834	8.08
Total:	12.740	52.747	4.70	32.052	197.010	6.35

Table 4 Bus 2(Cable): Comparison of Load Point IEAR Using the BIM and the CEM.

Load point	Failure rate (1/yr)	r (h)	U (h/yr)	EENS (MWh/yr)	ECOST (BIM) (1000\$/yr)	IEAR (BIM) (\$/kWh)	IEAR (CEM) (\$/kWh)	%Diff.
1	0.153	13.22	2.02	1.082	2.979	2.75	3.94	30.20
2	0.161	14.05	2.26	1.210	3.471	2.87	4.01	28.43
3	0.161	14.05	2.26	1.210	3.471	2.87	4.01	28.43
4	0.153	13.22	2.02	1.144	6.125	5.35	9.42	43.20
5	0.161	14.05	2.26	1.280	7.276	5.68	9.63	41.02
6	0.159	13.85	2.20	1.000	12.965	12.97	15.81	17.96
7	0.161	13.04	2.10	0.953	12.062	12.65	15.63	19.06
8	0.086	22.47	1.93	1.932	15.726	8.14	8.43	3.44
9	0.086	20.58	1.77	2.036	16.348	8.03	8.41	4.51
10	0.155	13.08	2.03	1.085	2.964	2.73	3.93	30.53
11	0.161	14.05	2.26	1.210	3.471	2.87	4.02	28.60
12	0.163	14.25	2.32	1.045	3.027	2.90	4.03	28.04
13	0.161	12.71	2.05	1.158	5.960	5.15	9.27	44.44
14	0.163	12.92	2.11	1.192	6.234	5.23	9.33	43.94
15	0.155	13.08	2.03	0.921	11.662	12.67	15.64	18.99
16	0.161	14.05	2.26	1.027	13.396	13.04	15.86	17.78
17	0.155	13.43	2.08	0.937	2.606	2.78	3.97	29.97
18	0.155	13.08	2.03	0.913	2.493	2.73	3.93	30.53
19	0.163	13.91	2.27	1.021	2.907	2.85	4.01	28.92
20	0.163	13.91	2.27	1.284	7.221	5.62	9.61	41.52
21	0.161	12.71	2.05	1.158	5.960	5.15	9.27	44.44
22	0.163	12.92	2.11	0.936	12.054	12.61	15.60	19.17

Table 5 Buses 2&4: System Performance Indices.

Index	2 (OHL)	2 (Cable)	4 (OHL)	4 (Cable)
SAIFI	0.248	0.158	0.300	0.190
SAIDI	0.765	2.169	0.621	1.449
CAIDI	3.084	13.686	2.071	7.619

Table 6 Buses 2&4: Composite Customer Damage Function.

System	Interrupted Cost (\$/kW)				
	1 min.	20 min.	1 hr	4 hr	8 hr
Bus 2	0.3610	1.3620	3.7564	13.7866	38.0000
Bus 4	0.7074	1.9692	4.9357	16.2724	39.9674

Table 7 Buses 2&4: Comparison of Bus IEAR Using the SIM and the CEM.

Bus (Case)	CAIDI (h/cus-int)	Cost of CAIDI (\$/kW)	IEAR (SIM) (\$/kWh)	IEAR (CEM) (\$/kWh)	%Difference
2(OHL)	3.084	10.803	3.50	3.94	11.17
2(Cable)	13.686	83.340	6.09	8.43	27.76
4(OHL)	2.071	9.235	4.46	4.70	5.11
4(Cable)	7.619	37.516	4.92	6.35	22.52

Table 8 Buses 2&4: Summary of Bus IEAR Using the Three Methods.

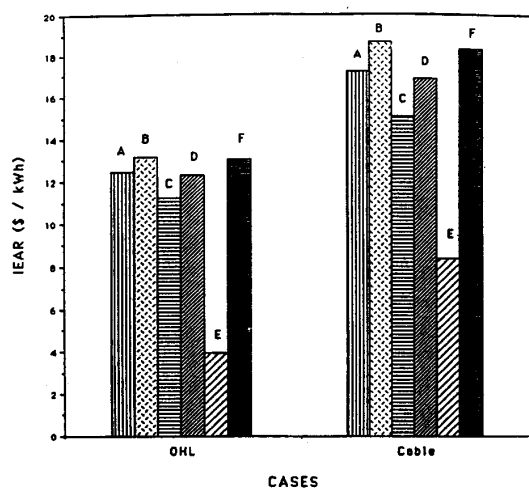
Method	2 (OHL)	2 (Cable)	4 (OHL)	4 (Cable)
CEM	3.94	8.43	4.70	6.35
BIM	3.41(13.4)	6.28(25.5)	4.44(5.5)	5.00(21.2)
SIM	3.50(11.2)	6.09(27.8)	4.46(5.1)	4.92(22.5)

Note: The numbers in parentheses are the percentage differences in reference to the CEM.

Table 9 Impact of Different System Configurations: Comparison of IEAR Using the Three Methods.

Case	CEM	OHL Case BIM	SIM	CEM	Cable Case BIM	SIM
BUS 2:						
A	12.46	6.23(50.0)	6.26(49.7)	17.20	9.99(41.9)	8.97(47.8)
B	13.14	10.87(17.3)	9.69(26.2)	18.64	17.14(8.0)	13.09(29.8)
C	11.27	6.76(40.0)	6.69(40.6)	15.06	12.78(15.1)	10.68(29.1)
D	12.34	5.36(56.6)	5.55(55.0)	16.88	7.91(53.1)	7.60(54.9)
E	3.94	3.41(13.4)	3.50(11.2)	8.43	6.28(25.5)	6.09(27.7)
F	13.09	8.87(32.2)	6.64(49.3)	18.33	13.78(24.8)	9.02(50.8)
BUS 4:						
A	8.72	5.72(34.4)	5.57(36.1)	10.61	6.65(37.3)	6.79(36.0)
B	8.42	7.48(11.2)	7.81(7.2)	10.96	10.45(4.6)	9.51(13.2)
C	7.48	5.63(24.7)	5.99(20.0)	9.52	8.81(7.4)	8.24(13.4)
D	8.67	5.52(36.3)	4.99(42.4)	10.12	5.96(41.1)	6.03(40.4)
E	4.70	4.44(5.5)	4.46(5.1)	6.35	5.00(21.2)	4.92(22.5)
F	8.36	6.56(21.5)	6.38(23.7)	10.58	8.73(17.5)	7.83(26.0)

Note: The numbers in parentheses are the percentage differences in reference to the CEM.



Case A : Disconnects; fuses; alternate supply; transformer repair.

Case B : No disconnects; no fuses; no alternative supply; transformer repair.

Case C : No disconnects; fuses; no alternate supply; transformer repair.

Case D : Disconnects; no fuses; alternative supply; transformer repair.

Case E (Base case): Disconnects; fuses; alternative supply; transformer replacement.

Case F : Disconnects; no fuses; no alternative supply; transformer repair.

Figure 3 Bus 2: Effects on the Bus IEAR of Different Configurations (CEM used).

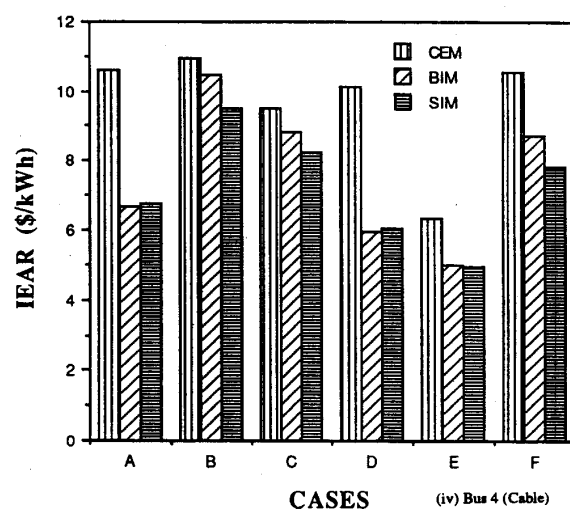
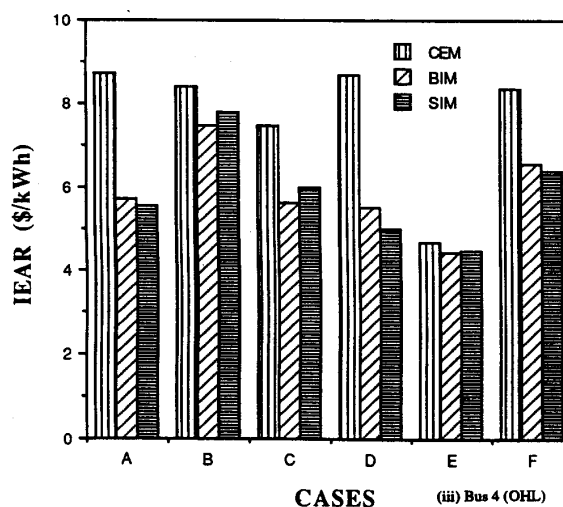
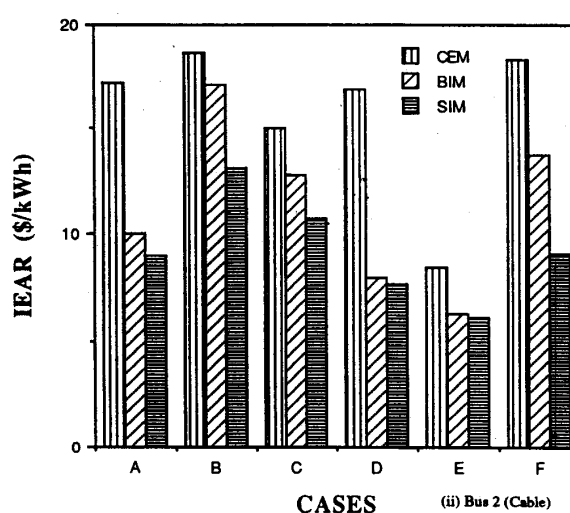
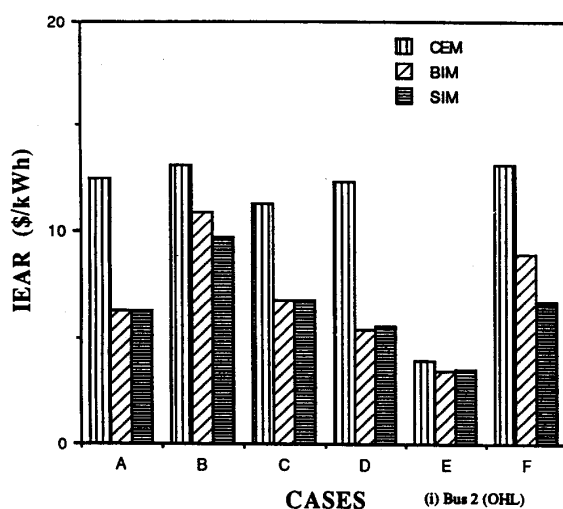


Figure 4 Comparison of the Bus IEAR using the Three Methods (i) Bus 2 (OHL) (ii) Bus 2 (Cable) (iii) Bus 4 (OHL) (iv) Bus 4 (Cable).